

Calculator -> react cypress

1. npx create-react-app cypressfol
2. cd cypressfol
3. npm install --save-dev cypress
4. Create cypress directory in cypressfol => ./cypressfol/cypress
5. create e2e directory in cypress => ./cypressfol/cypress/e2e

here create calculator.cy.js

calculator.cy.js

```
// cypress/e2e/app.cy.js
describe('Calculator E2E Test', () => {
  beforeEach(() => {
    cy.visit('http://localhost:3000'); // app must be running
  });

  it('shows calculator title', () => {
    cy.contains('React Calculator Example').should('be.visible');
    cy.contains('Calculator').should('be.visible');
  });

  it('adds two numbers', () => {
    cy.get('[data-testid="numA"]').type('8');
    cy.get('[data-testid="numB"]').type('2');
    cy.get('[data-testid="addBtn"]').click();
    cy.get('[data-testid="result"]').should('have.text', 'Result: 10');
  });

  it('subtracts two numbers', () => {
    cy.get('[data-testid="numA"]').type('8');
    cy.get('[data-testid="numB"]').type('2');
    cy.get('[data-testid="subBtn"]').click();
    cy.get('[data-testid="result"]').should('have.text', 'Result: 6');
  });

  it('multiplies two numbers', () => {
    cy.get('[data-testid="numA"]').type('6');
    cy.get('[data-testid="numB"]').type('7');
    cy.get('[data-testid="mulBtn"]').click();
    cy.get('[data-testid="result"]').should('have.text', 'Result: 42');
  });

  it('divides two numbers', () => {
    cy.get('[data-testid="numA"]').type('20');
    cy.get('[data-testid="numB"]').type('4');
    cy.get('[data-testid="divBtn"]').click();
    cy.get('[data-testid="result"]').should('have.text', 'Result: 5');
  });

  it('shows error when dividing by zero', () => {
    cy.get('[data-testid="numA"]').type('10');
    cy.get('[data-testid="numB"]').type('0');
    cy.get('[data-testid="divBtn"]').click();
  });
});
```

```

    cy.get('[data-testid="result"]').should('have.text', 'Result: Error');
  });
  it('clears inputs and result', () => {
    cy.get('[data-testid="numA"]').type('9');
    cy.get('[data-testid="numB"]').type('3');
    cy.get('[data-testid="addBtn"]').click();
    cy.get('[data-testid="clearBtn"]').click();
    cy.get('[data-testid="numA"]').should('have.value', '');
    cy.get('[data-testid="numB"]').should('have.value', '');
    cy.get('[data-testid="result"]').should('have.text', 'Result: -');
  });
});
});

```

6. create fixtures directory in cypressfol => ./cypressfol/fixtures

here create support directory => ./cypressfol/fixtures/support

now in support create e2e.js and example.json

e2e.js

```

// Cypress support file
import './commands';

```

example.json

```

{
  "name": "Abhi"
}

```

7. In src create components directory => ./src/components

here in components Calculator.js and Calculator.test.js

Calculator.js

```

// components/Calculator.js
import React, { useState } from 'react';

const Calculator = () => {
  const [numA, setNumA] = useState('');
  const [numB, setNumB] = useState('');
  const [result, setResult] = useState(null);

  const handleOperation = (type) => {
    const a = Number(numA);
    const b = Number(numB);

    switch (type) {
      case 'add':
        setResult(a + b);
    }
  };
}

```

```

        break;
      case 'sub':
        setResult(a - b);
        break;
      case 'mul':
        setResult(a * b);
        break;
      case 'div':
        setResult(b !== 0 ? a / b : 'Error');
        break;
      default:
        setResult(null);
    }
  };

  const handleClear = () => {
    setNumA('');
    setNumB('');
    setResult(null);
  };

  return (
    <div>
      <h2>Calculator</h2>
      <input
        type="number"
        data-testid="numA"
        value={numA}
        onChange={(e) => setNumA(e.target.value)}
        placeholder="Enter first number"
      />
      <input
        type="number"
        data-testid="numB"
        value={numB}
        onChange={(e) => setNumB(e.target.value)}
        placeholder="Enter second number"
      />
      <div>
        <button data-testid="addBtn" onClick={() =>
handleOperation('add')}>Add</button>
        <button data-testid="subBtn" onClick={() =>
handleOperation('sub')}>Subtract</button>
        <button data-testid="mulBtn" onClick={() =>
handleOperation('mul')}>Multiply</button>
        <button data-testid="divBtn" onClick={() =>
handleOperation('div')}>Divide</button>
        <button data-testid="clearBtn" onClick={handleClear}>Clear</button>
      </div>
      <p data-testid="result">
        Result: {result !== null ? result : '-'}
      </p>
    </div>
  );
};

export default Calculator;

```

Calculator.test.js

```
// components/Calculator.test.js
import { render, screen, fireEvent } from '@testing-library/react';
import Calculator from './Calculator';
import '@testing-library/jest-dom';

test('renders calculator title', () => {
  render(<Calculator />);
  expect(screen.getByText('Calculator')).toBeInTheDocument();
});

test('adds two numbers', () => {
  render(<Calculator />);
  fireEvent.change(screen.getByTestId('numA'), { target: { value: '10' } });
  fireEvent.change(screen.getByTestId('numB'), { target: { value: '5' } });
  fireEvent.click(screen.getByTestId('addBtn'));
  expect(screen.getByTestId('result')).toHaveTextContent('Result: 15');
});

test('subtracts two numbers', () => {
  render(<Calculator />);
  fireEvent.change(screen.getByTestId('numA'), { target: { value: '10' } });
  fireEvent.change(screen.getByTestId('numB'), { target: { value: '5' } });
  fireEvent.click(screen.getByTestId('subBtn'));
  expect(screen.getByTestId('result')).toHaveTextContent('Result: 5');
});

test('multiplies two numbers', () => {
  render(<Calculator />);
  fireEvent.change(screen.getByTestId('numA'), { target: { value: '4' } });
  fireEvent.change(screen.getByTestId('numB'), { target: { value: '3' } });
  fireEvent.click(screen.getByTestId('mulBtn'));
  expect(screen.getByTestId('result')).toHaveTextContent('Result: 12');
});

test('divides two numbers', () => {
  render(<Calculator />);
  fireEvent.change(screen.getByTestId('numA'), { target: { value: '20' } });
  fireEvent.change(screen.getByTestId('numB'), { target: { value: '4' } });
  fireEvent.click(screen.getByTestId('divBtn'));
  expect(screen.getByTestId('result')).toHaveTextContent('Result: 5');
});

test('handles divide by zero', () => {
  render(<Calculator />);
  fireEvent.change(screen.getByTestId('numA'), { target: { value: '10' } });
  fireEvent.change(screen.getByTestId('numB'), { target: { value: '0' } });
  fireEvent.click(screen.getByTestId('divBtn'));
  expect(screen.getByTestId('result')).toHaveTextContent('Result: Error');
});

test('clears inputs and result', () => {
  render(<Calculator />);
  fireEvent.change(screen.getByTestId('numA'), { target: { value: '9' } });
  fireEvent.change(screen.getByTestId('numB'), { target: { value: '3' } });
  fireEvent.click(screen.getByTestId('clearBtn'));
  expect(screen.getByTestId('numA')).toHaveTextContent('');
  expect(screen.getByTestId('numB')).toHaveTextContent('');
  expect(screen.getByTestId('result')).toHaveTextContent('');
});
```

```

    fireEvent.click(screen.getByTestId('addBtn'));
    fireEvent.click(screen.getByTestId('clearBtn'));
    expect(screen.getByTestId('numA')).toHaveValue(null);
    expect(screen.getByTestId('numB')).toHaveValue(null);
    expect(screen.getByTestId('result')).toHaveTextContent('Result: -');
  });
};

```

8. App.js

```

// App.js
import React from 'react';
import Calculator from './components/Calculator';

function App() {
  return (
    <div>
      <h1>React Calculator Example</h1>
      <Calculator />
    </div>
  );
}

export default App;

```

9. App.test.js

```

// App.test.js
import { render, screen } from '@testing-library/react';
import App from './App';
import '@testing-library/jest-dom';

test('renders app title', () => {
  render(<App />);
  expect(screen.getByText('React Calculator Example')).toBeInTheDocument();
});

test('renders calculator heading', () => {
  render(<App />);
  expect(screen.getByText('Calculator')).toBeInTheDocument();
});

test('shows initial result placeholder', () => {
  render(<App />);
  expect(screen.getByTestId('result')).toHaveTextContent('Result: -');
});

```

10. index.js

```
import React from 'react';
import ReactDOM from 'react-dom/client';
import './index.css';
import App from './App';

const root = ReactDOM.createRoot(document.getElementById('root'));
root.render(
  <React.StrictMode>
    <App />
  </React.StrictMode>
);
```

11. Then type npm test and npm start

12. In second local terminal type

cd cypressfol

npm run cypress:open